METHOD:

Ethics and Study Design: Recruitment Procedure:

This single-centre study was performed at a tertiary hospital in Australia, and was approved by the local Human Research and Ethics Committee in accordance with the requirements of the National Statement on Ethical Conduct in Human Research.

Study participants were asked to complete three surveys over the course of the study, each survey containing the same select MRI images from 25 MRI lumbar spine cases. The surveys were delivered at 0 (Baseline), 1 and 6 weeks from study commencement. For each case, participants were asked to grade the degree of foraminal stenosis on a scale from 0 to 3 (0 = no stenosis, 1 = mild stenosis, 2 = moderate stenosis and 3 = severe stenosis). Following completion of the baseline survey, participants were asked to read an educational document summarising the grading scale developed by Lee et al (7).

Recruitment Procedure:

All MRI reporting consultant radiologists and registrars were invited to participate. This was defined as those rostered to MRI reporting at least once per month. A total of 12 participants provided their written consent to participate in the study.

Image Selection / Sample Size:

Select sagittal T1 and T2 images were obtained via retrospective review of lumbar spine MRI studies performed at Barwon Health, with a total of twenty-five studies selected. The estimated optimal number of studies to be examined was calculated by assuming that at least 6 raters completed the study. Assuming the percentage agreement between raters has an 80% error margin, a minimum of 25 MRI studies was required.

The studies were acquired on either a Siemens Magnetom Prisma 3 Tesla, or Siemens Magnetom Avanto Fit 1.5 Telsa MRI machine. Images were de- identified.

Data Collection and Analysis:

Each survey was delivered using the REDCap program. Data was collected using the REDCap Longitudinal Survey, with participants survey results anonymous. Data was obtained at baseline prior to participants being provided the education material summarising the Lee et al (7) grading scale, then 1 week and 6 weeks later.

Survey results were analysed using the Stata package, Kappatec. Categorical descriptive data was assessed using frequencies and percentages. Inter-rater agreement was assessed using Gwet's coefficient and Krippendoff's alpha, reported together with their 95% confidence interval. Agreement coefficients of the 1 week and 6 week post intervention were compared with the baseline coefficient using Gwet's paired T-test. Gwet's method was used instead of the commonly used Cohens Kappa, as it allows comparison of agreement at baseline and following an intervention. A p-value of <0.05 was considered statistically significant.